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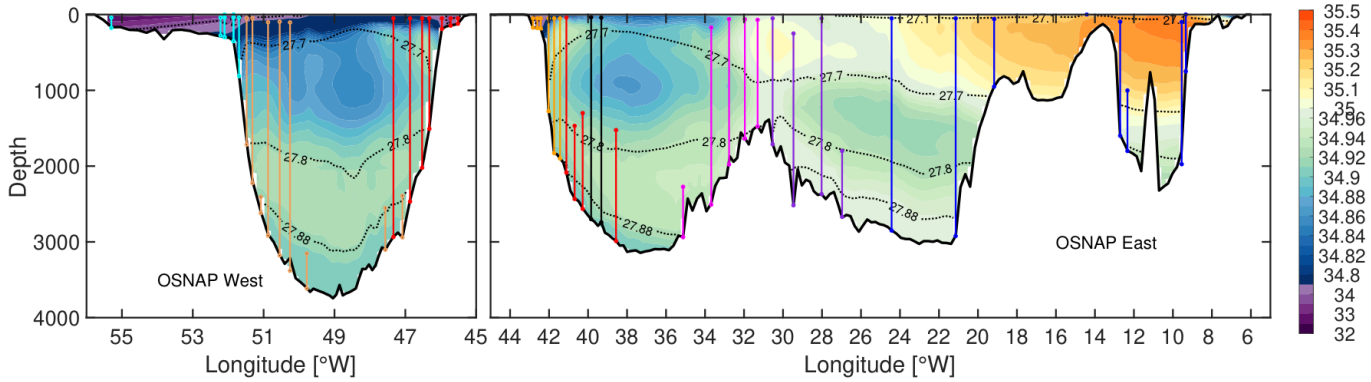
Week 1 - Choosing regions, timescales and topics

Put your name against 4 topics below: after the "Students interested"
Bold your top choices
For students who don't need credit, put your name against 1-2 as "Moderator"

- EFW/JB - underline presenters



Topics

List of topics (~~not in order~~ - now in order)**:**

1. **<u>What drives the AMOC?</u>**
   Southern ocean winds, mixing, diabatic vs adiabatic
   - Gnanadesikan (1999): Simple conceptual model
   - Toggweiler & Samuels (1995): Southern ocean wind stress
   - Cessi & Wolfe (2009): Adiabatic AMOC
   - See also other references in Johnson et al. (2019)
   - Students interested: **<u>Emelie</u>**, Andrea, **<u>Sreya</u>**, Jordis, Susie
   - Moderator: María
2. **<u>South Atlantic Dynamics: The Gateway</u>**
   How important is the South Atlantic for the AMOC?
   - Bryden (obs) or FOV theory (?)
   - Biastoch et al.: Agulhas leakage and stability
   - Recent reviews ..
   - Students interested: **<u>Tali</u>**, Andrea, **<u>Zubair</u>**, Lea
   - Moderator:

   Deep water formation: Myth, measurement & mechanism
   How do overflows and convection shape deep water formation?

- Osterhus et al. (2019) - stability of overflows
- Lozier (2012) - disconnect between convection and overturning
- Students interested: Tali, Emelie, Arran, Andrea
- Moderator: Isabelle

3. **Observing the Overturning: From snapshots to time series**
   How do we measure AMOC transport, and what have we learned?
   - Bryden et al. (2005): hydrographic evidence for slowing AMOC
   - Kanzow et al. (2011)/Chidichimo et al. (2010): Temporal aliasing and seasonal cycles
   - Ganachaud & Wunsch (2003): Inverse approaches from hydrographic data
   - Frajka-Williams et al. (2018): Geostrophic reference levels - obs
   - Danabasoglu et al. (2021): Geostrophic reference levels - models
   - Students interested: **Arran**, Jan, Jordis, Arnaz, Zubair
   - Moderator:

   Indirect approaches: Satellites, Reanalyses and Budgets
   How do other AMOC estimates compare?
   - Satellite-derived transport estimates: Frajka-Williams (2015), Willis (2011), Sanchez-Franks (2021), Le Bras et al. (2023)
   - Karspeck et al. (2019), Jackson et al. (2022), Baehr et al. (2015): Ocean reanalyses
   - Calafat et al. (2025)/Kelly et al. (2016): Budgetary approach to heat transport
   - Students interested: Arran, **Jan**, Andreas, Zubair, Arnaz
   - Moderator:

4. **Connectivity and Adjustment: Linking the Atlantic**
   How does the AMOC communicate along its path?
   - Johnson and Marshall (2002): Adjustment processes
   - Zhang (2010): Time-lagged signals between the subpolar and subtropical AMOC
   - Van Sebille et al. (2011) /Le Bras et al. (2017): Passive tracers (spice) with time lag from observations.
   - Students interested: Tali, Emelie, Andreas, Sreya, Lea, **Susie**, Jan
   - Moderator: María

5. **Fingerprints of Change: SSTs and the AMV**
   Can we detect the AMOC change in surface signals?
   - Caesar et al. (2018): SST fingerprint of the AMOC
   - Terhaar et al. (2024): Alternate interpretations for SST / heat flux
   - Students interested: Jan, **Jordis**, **Arnaz**, Andreas, Zubair, Susie
   - Moderators: Isabelle

6. **Tipping Points and Paleo Perspectives**
   What do models and paleo data say about AMOC thresholds?
   - Weijer et al. (2019): Review of tipping and hosing experiments
   - Broecker (1991): Thermohaline conveyor shutdowns
   - Boers / Ditlevsen / Baker / van Weston: Recent tipping point debates
   - Students interested: **Andreas**, **Lea**, Jordis, Susie, Arran
   - Moderators: Keno

7. **Simple models: Foundations of overturning theory**
   How do idealised models help us understand AMOC behaviour?
   - Stommel (1961): Box model and multiple equilibria
   - Sandström (horizontal convection) or Welander loop models
   - See also introduction in Johnson et al. (2019) for other papers.
   - Students interested: Tali, Emelie, Sreya, Lea, Arnaz, **Andrea, Lara**
   - Moderator: Keno

What do Models Say? (IPCC & Beyond)
What do climate models project for the AMOC and its **impacts**?

- IPCC AR6 chapter excerpts
- Selected CMIP6 studies
- Students: This is a topic for week 11 when all students present something for their region.

People who aren't enrolled for credit (moderators): Isabelle, Maria-Jesus, Isabel, Keno

Review papers

Some additional references - these are review style and *not* a good choice for the planned paper discussion, but may help you either find references or also place them in context.

Introduction to a review series from 2019:

- Srokosz, Danabasoglu, Patterson (2021) AMOC: Reviews of observational and modelling advances–An introduction. JGR Oceans. http://doi.org/10.1029/2020JC016745

Some of the papers in the review series that may be relevant (use them to find individual papers - you should not use a review for the paper discussion):

- Frajka-Williams et al. (2019) AMOC: Observed transport and variability. Frontiers. https://doi.org/10.3389/fmars.2019.00260
- Hirschi et al. (2020) The AMOC in high resolution models. JGR Oceans. https://doi.org/10.1029/2019JC015522
- Johnson et al. (2019) Recent contributions of theory to our understanding of the AMOC. JGR Oceans. https://doi.org/10.1029/2019JC015330
- Weijer et al. (2019) Stability of the AMOC: A review and synthesis. JGR Oceans. https://doi.org/10.1029/2019JC015083
- Zhang et al. (2019) A review of the role of the AMOC in Atlantic multidecadal variability and associated climate impacts. *Reviews of Geophysics.* https://doi.org/10.1029/2019RG000644

Schedule

Weekly seminar: Tuesdays, 14:15-15:45 Bu53 133

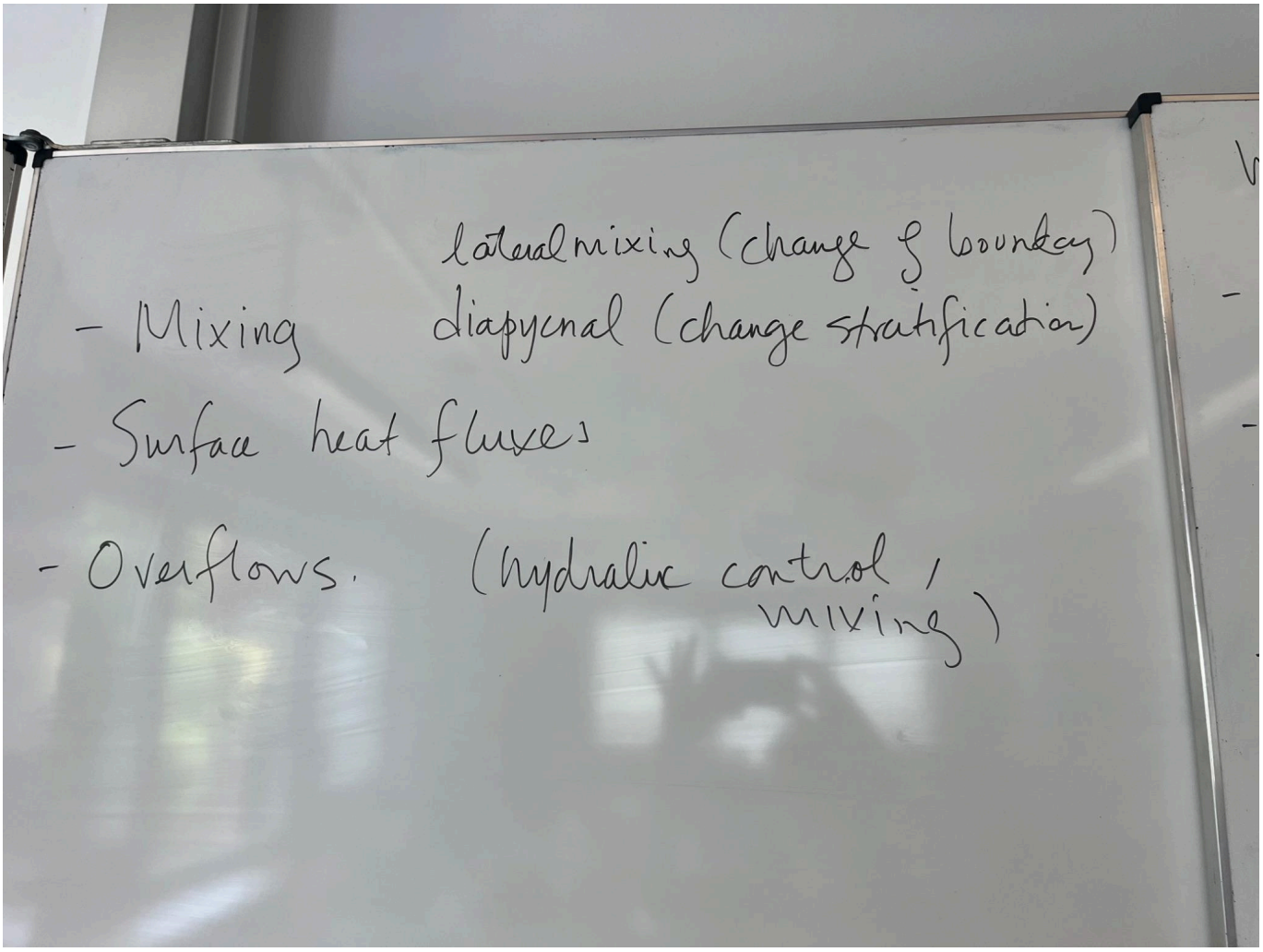
	Date	Topic	Notes		EFW/JB
1	8 Apr	Overview			
2	15 Apr	What is the AMOC?			JB - seminar 3:15
3	22 Apr	Observing			
		Topic	Presenter 1	Presenter 2	
4	29 Apr	What drives the AMOC? (Maria)	Sreya - Gnanadesikan "A Simple Predictive Model for the Structure of the Oceanic Pycnocline"	Emelie - Toggweiler and Samuels 1995	JB - EGU Andrea/Lara /Keno
5	6 May	South Atlantic (*Maria*)	Tali - Bryden ethttps://drive.google.com/file/d/1FUcn3ziNCSj-bR245gflTCcpxmlS9exc/view?usp=sharing al (2011) https://drive.google.com/drive/folders/1Xv_cUrsvxq6i6hWZ55ydD9v89L50XCLK	Zubair - Beal et. al (2011) https://doi.org/10.1038/nature09983	
6	13 May	Observing / Indirect (Isabelle)	Arran - Kanzow et al, (2010) arranbolger@gmail....	Frajka-Williams,2015. (jan.barcelona.martin@studium.uni-hamburg.de)	JB - away Lara Emelie Jordis
7	20 May	Connectivity (Maria)	Susie - Zhang (2010)	What processes are important, and how well can we observe or model them?	Andrea away
	27 May	Pfingsten	No class scheduled		
8	3 Jun	Fingerprints (Isabelle)	Jordis - Terhaar et al. 2025 Papers or https://doi.org/10.1038/s41467-024-55297-5	What processes are important, and how well can we observe ~~or model them~~? **Top 3 processes** most important/informative for present day AMOC (which of these would you observe for the most informed statement about the AMOC)	JB - away? Tali away Lea away Andrea away ? Susie away

9	10 Jun				JB/EFW - away
10	17 Jun	Tipping points (Keno)	Andreas - https://www.nature.com/articles/s41558-021-01097-4	Lea https://www.nature.com/articles/s41586-024-08544-0	EFW - away Emelie
11	24 Jun	Simple models (—)	Andrea - Welander (1982)	Lara – Stommel (1961)	
XX	30 Jun		Eleanor's seminar: 30th June at 11:00am "Changing Currents: 2 decades of AMOC obs" https://geomar.webex.com/geomar-en/j.php?MTID=mf415834ed76a742bacf01eada863885d		
12	1 Jul	Impacts	Each student adds one figure to "impacts" slide deck to present. **Recommended:** Find the IPCC AR6 report where it talks about the AMOC (possibly chapter 9). Use this to find a reference(s) about CMIP models looking at consequences of AMOC change.		
13	8 Jul	past/ present/ future AMOC	**What did you learn/ do you know about the AMOC**: - Past state - Present state - Future Evolution Please provide one sentence each (inkl. a reference) over here by **Monday**: https://docs.google.com/presentation/d/14fUIT_NhIk4yI0ibc3cZiMZYGVKMOqAYq5CCBP8OWuU/edit?usp=sharing You may do this from your focus point/region or the general perspective, that is, what you learned in the course		EFW - away
14	15 Jul	feedback			Lara away IMPRS

GitHub username

1. Tali - blaueinhalte
2. Susie: susiebrn
3. María: maryrapanague
4. Arran: ArranBolger
5. Lea: lea-dietz
6. Zubair : zubairasyraf169
7. Isabel: IsabelAC0409
8. Jordis: jordis-hansen
9. Emelie: EmelieOcean
10. Sreya: sreyaps
11. Jan: Jan-BCN
12. Andrea: andreamosso
13. Andreas: dre1724
14. Keno: kriechers
15. Lara: larah313

20th May - Processes for the AMOC



What processes are impt (for AMOC)

- Agulhas leakage
- Wind everywhere — tradewinds
  Ekman — Southern ocean
  — northern North Atl
  gyre ← wind stress curl
- NADW formation
  watermass transformation
- Ice sheet melting + formation
  - freshwater fluxes / evaporation
- (Boundary) wave adjustment
- Advection of anomalies in boundary
  density / dyn height